



PaySim Fraud Detection & Risk Analysis

The domain of the Project: Data Analytics & Artificial Intelligence

Course : Data Analytics

Team Mentors (and their designation):

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Period of the project

July 2026 to September 2026



Declaration

The project titled **PaySim Fraud Detection & Risk Analysis** has been undertaken as part of the Data Analytics program organized by SURE Trust - IERY. The project applies data analytics, exploratory analysis, supervised machine learning, fraud-risk analysis, and interactive application development to financial transaction data.

I declare that, to the best of my knowledge, the work presented in this report reflects the analysis and implementation carried out for the project. The report documents the PaySim data preparation, exploratory analysis, machine-learning model development, model evaluation, dashboard work, and Streamlit-based fraud detection interface.

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1. Executive Summary

- The PaySim Fraud Detection & Risk Analysis project focuses on using transaction-level data to identify suspicious financial activity and support faster, more scalable fraud monitoring. The project combines data preparation, exploratory data analysis, feature engineering, supervised machine learning, visualization, and an interactive Streamlit interface.
- The prepared PaySim dataset contains 6,362,620 transaction records and 20 columns. It contains 8,213 fraudulent transactions, giving a fraud rate of 0.1291%, which makes the problem strongly imbalanced. No missing values were reported in the notebook analysis. The very small fraud class makes recall, precision, F1-score, and the confusion matrix more informative than accuracy alone.
- Exploratory analysis found that fraudulent activity is concentrated in TRANSFER and CASH_OUT transactions in the prepared data. The observed fraud rate was approximately 0.7688% for TRANSFER and 0.1840% for CASH_OUT, while PAYMENT, CASH_IN, and DEBIT had no recorded fraud cases in the analysis output. Transaction amount was highly skewed, with a maximum transaction amount of 92,445,516.
- The machine-learning implementation uses a preprocessing-and-model pipeline. Numeric transaction variables are standardized, the transaction type is one-hot encoded, and Logistic Regression with `class_weight="balanced"` is trained for binary classification. The training process uses a 70/30 stratified train-test split so the rare fraud class is represented in both partitions.
- On the recorded test results, the model achieved 94.54% accuracy and 94% recall for fraudulent transactions. However, fraud precision was only 2% and fraud F1-score was 4%, with 104,076 legitimate transactions incorrectly flagged as fraud. This means the model is useful as an exploratory high-recall baseline but would require threshold tuning, stronger models, and false-positive reduction before production use.
- An interactive Streamlit application was also developed. The project contains a separate dashboard for overview, fraud analysis, and risk/account analysis, as well as a transaction input interface. The current transaction input interface includes manually defined risk rules and should be clearly treated as a prototype unless connected directly to the saved ML pipeline.

Key takeaway: The project demonstrates the complete analytics-to-application workflow, while the model evaluation highlights the central fraud-detection trade-off between catching more fraud and avoiding excessive false alerts.



2. Introduction

2.1 Background and Context of the Project

Digital payment systems generate large volumes of transactions, making manual review difficult to scale. Fraud monitoring therefore benefits from systematic analysis of transaction characteristics, account balances, transaction types, and historical fraud labels.

PaySim provides a practical environment for studying this problem because the dataset includes transaction amount, transaction type, sender and receiver balances, and fraud indicators. The project uses these variables to understand normal versus suspicious transaction patterns and to build a classification baseline.

The project is not intended to replace operational fraud teams. Instead, it demonstrates how an analytics solution can prioritize transactions for review and give users an accessible way to explore transaction patterns.

2.2 Problem Statement

How can transaction data be analyzed and used to identify potentially fraudulent transactions in an efficient and user-friendly manner?

A useful fraud-detection solution must balance two competing risks: missing fraudulent activity and generating too many false alerts on legitimate transactions. This project therefore treats fraud detection as an imbalanced binary classification problem and evaluates the model with metrics that expose this trade-off.

2.3 Scope of the Project

- Understand the PaySim transaction schema and fraud labels.
- Prepare and inspect transaction data for quality issues and missing values.
- Perform exploratory analysis of transaction type, transaction amount, balances, time, and fraud patterns.
- Create analytical features such as sender/receiver balance differences and other derived indicators.
- Train and evaluate a supervised Logistic Regression baseline with balanced class weights.
- Build interactive Streamlit dashboards and a transaction-level fraud checking interface.
- Document the workflow, findings, limitations, and future improvements.

2.4 Limitations of the Project

- The PaySim dataset is simulated and may not capture every characteristic of production banking data.



- The current saved ML model is a Logistic Regression baseline and produces many false positives at the recorded decision threshold.



- ROC-AUC and PR-AUC were not reported in the supplied notebook outputs, so they are not claimed in this report.
- The transaction input screen in the supplied application uses rule-based scoring logic and should not be described as a probability-producing ML interface.
- A production deployment would require monitoring, data drift checks, threshold governance, security controls, explainability, and continuous validation.

2.5 Innovation Component

The innovation component is the combination of analytical fraud investigation with an interactive user-facing application. The project moves from raw transaction data to a model artifact and then to a Streamlit interface, making the result easier to demonstrate and use than a notebook-only solution.



3. Project Objectives

- 1. Understand the dataset:** Inspect the PaySim structure, labels, transaction types, balances, and data quality.
- 2. Prepare transaction data:** Create and validate derived variables needed for analysis and modelling.
- 3. Perform EDA:** Study transaction amounts, transaction types, fraud rates, balance behavior, and fraud over time.
- 4. Engineer model features:** Select usable predictors and remove identifier fields from the ML baseline.
- 5. Train a classification model:** Build a reusable preprocessing + Logistic Regression pipeline with balanced class weights.
- 6. Evaluate the model:** Use classification report, confusion matrix, accuracy, precision, recall, and F1-score.
- 7. Develop an interactive application:** Present analytics through Streamlit and provide a transaction-level fraud assessment interface.
- 8. Document learnings:** Capture the business problem, technical workflow, results, limitations, and future scope.

3.1 Expected Outcomes and Deliverables

- Prepared PaySim analysis dataset and profiling results
- EDA findings and fraud pattern analysis
- Machine-learning training pipeline and saved model artifact
- Model evaluation results and confusion matrix
- Interactive Streamlit dashboard
- Fraud-detection input application
- README and project report



4. Methodology and Results

4.1 Dataset Understanding

Measure	Observed result
Rows	6,362,620
Columns	20
Fraud transactions	8,213
Legitimate transactions	6,354,407
Fraud rate	0.1291%
Missing values	0
isFlaggedFraud = 1	16
Transaction amount min / max	0 / 92,445,516

4.2 Important Transaction Variables

Variable	Role in the project
type	Transaction category used as a categorical predictor.
amount	Transaction value; highly skewed and ranged up to 92,445,516 in the supplied analysis.
oldbalanceOrg / newbalanceOrig	Sender balance before and after the transaction.
oldbalanceDest / newbalanceDest	Receiver balance before and after the transaction.
isFraud	Target variable: 0 = legitimate, 1 = fraudulent.
isFlaggedFraud	Original fraud flag indicator; removed from the ML feature set in the notebook.



step / hour / day	Time-related variables used during analysis; step was later removed before modelling.
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Variable	Role in the project
nameOrig / nameDest	Account identifiers used for exploratory analysis and then removed from the ML baseline.

4.3 Data Preparation and Exploratory Analysis

- The supplied notebook checked dataset shape, fraud-class counts, flagged-fraud counts, missing values, and basic descriptive statistics.
- Additional analytical variables included balance differences for sender and receiver accounts. The notebook also created or used time-related fields such as hour, day, and time period for trend analysis.
- The analysis observed that fraud was associated with TRANSFER and CASH_OUT in the recorded data. Fraud rates were 0.7688% for TRANSFER and 0.1840% for CASH_OUT; the remaining transaction types had 0 fraud cases in the output shown in the notebook.
- A time-based chart was used to inspect fraud incidents across transaction steps, and a correlation matrix was produced for amount, balances, and the fraud label.
- The analysis also counted transactions where the sender balance became zero after a TRANSFER or CASH_OUT. This pattern was treated as useful context for understanding suspicious transaction behavior, not as proof of fraud by itself.

4.4 Fraud Pattern Findings

Transaction type	Fraud-rate value	Approx. percent
TRANSFER	0.007688	0.7688%
CASH_OUT	0.001840	0.1840%
CASH_IN	0	0%
DEBIT	0	0%
PAYMENT	0	0%

Imbalance warning: Only 0.1291% of all transactions are fraudulent in the prepared dataset. A model can therefore appear accurate while still performing poorly on the fraud class.

4.5 Machine-Learning Approach



- Problem definition: binary classification of isFraud, where 0 represents legitimate transactions and 1 represents fraudulent transactions.
- Feature selection: nameOrig, nameDest, and isFlaggedFraud were removed from the modelling table. The remaining baseline predictors include transaction type and five numeric transaction/balance variables.
- Preprocessing: StandardScaler was applied to numeric variables, while OneHotEncoder(drop="first", handle_unknown="ignore") was applied to the transaction type.
- Model: LogisticRegression(class_weight="balanced", max_iter=1000). The class-weighting strategy was used to give the minority fraud class more influence during learning.
- Validation: the notebook used train_test_split with test_size=0.30 and stratify=y.

4.6 Model Evaluation Results

Metric	Recorded result
Accuracy	94.54%
Fraud precision	0.02 (2%)
Fraud recall	0.94 (94%)
Fraud F1-score	0.04 (4%)
Legitimate precision	1.00
Legitimate recall	0.95
Macro F1	0.51

4.7 Confusion Matrix

Actual class	Predicted legitimate	Predicted fraud
Actual legitimate	1,802,246	104,076
Actual fraud	153	2,311

Interpretation: The model catches most fraud cases (high recall) but also flags many legitimate transactions. The 104,076 false positives are the main reason precision and F1 for fraud are low.



4.8 Streamlit Dashboard

The dashboard implementation contains three main analytical areas: Executive Overview, Fraud Analysis, and Risk & Account Analysis. The interface supports transaction-type filtering and provides metrics and Plotly charts for transaction volume, fraud rate, fraud timing, amount/balance behavior, risk levels, and high-risk transaction inspection.

4.9 Fraud Detection Application

The supplied Streamlit prediction interface collects transaction type, amount, sender balances, and receiver balances. The current logic assigns a manually calculated risk score based on large amounts, high utilization of sender balance, sender-balance inconsistencies, and receiver-balance inconsistencies. The score is then mapped to SAFE / LEGITIMATE or FRAUDULENT based on a threshold of 50.

Figure: Streamlit fraud detection input interface supplied with the project.

4.10 Rule-Based Prototype Logic

Prototype rule	Effect
Amount $\geq$ 50,000	+30 risk points
TRANSFER and amount $\geq$ 80% of sender old balance	+30



CASH_OUT and amount $\geq$ 80% of sender old balance	+30
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Prototype rule	Effect
Sender balance inconsistency > 1	+20
Receiver balance inconsistency > 1 for TRANSFER/CASH_IN	+20
Final rule threshold	$\geq 50 \rightarrow$ FRAUDULENT; $< 50 \rightarrow$ SAFE / LEGITIMATE

Important labeling note: The prototype score is a heuristic risk score out of 100. It should not be labeled or interpreted as a calibrated probability of fraud.

4.11 Project Architecture

Stage	Implementation
1. Data	PaySim transaction records
2. Preparation	Quality checks, derived variables, feature selection
3. EDA	Transaction, amount, balance, type, and fraud analysis
4. ML	Preprocessing + Logistic Regression pipeline
5. Evaluation	Classification report + confusion matrix
6. Deployment layer	Saved pipeline artifact + Streamlit interface



5. Social/Industry Relevance of the Project

5.1 Industry Relevance

- Fraud detection is directly relevant to digital payments, banking, fintech, e-commerce, wallets, and other transaction-driven services.
- The project demonstrates an approach in which transaction data is converted into risk information that can support analyst review and prioritization.
- The use of a reusable ML pipeline shows how the same preprocessing used during training can be packaged with the model, reducing the risk of inconsistent transformation during deployment.

5.2 Social Relevance

- Reducing fraud can help protect customers from avoidable financial loss and improve trust in digital payment systems.
- Interactive dashboards can help non-programming users understand transaction activity and fraud patterns without reading model code.
- As a learning project, the work gives practical exposure to an important real-world data analytics problem and encourages responsible interpretation of model results.

5.3 Responsible Use

A fraud flag should be treated as a signal for further investigation, not as automatic proof of criminal behavior. Thresholds must be tested against business costs, and legitimate users should have a path for review when transactions are incorrectly flagged.



6. Learning and Reflection

6.1 Technical Learning

- Learned to inspect a large transaction dataset and verify its structure, class distribution, and data quality.
- Practiced EDA techniques including bar charts, distribution plots, box plots, time-based fraud analysis, and correlation analysis.
- Learned why imbalanced classification requires more than accuracy and how confusion matrices reveal the practical consequences of model decisions.
- Implemented preprocessing and model training as a single scikit-learn Pipeline so the transformation steps are preserved with the classifier.
- Gained experience connecting analytics to Streamlit so the project can be demonstrated interactively.

6.2 Key Insight from Model Evaluation

The most important modelling lesson was that high recall does not automatically mean a model is operationally good. The recorded Logistic Regression baseline found 2,311 of 2,464 fraud cases in the test set, but it also produced 104,076 false positives. This highlights the need to optimize fraud detection around business cost, not only headline accuracy or recall.

6.3 Areas for Improvement

- Calibrate and tune the fraud decision threshold.
- Compare additional models such as tree-based classifiers.
- Use precision-recall analysis and PR-AUC in addition to the existing metrics.
- Investigate more carefully engineered balance and transaction-behavior features.
- Integrate the saved ML pipeline directly into the Streamlit prediction page.
- Add model explainability and a review workflow for flagged transactions.



7. Conclusion and Future Scope

7.1 Conclusion

The PaySim Fraud Detection & Risk Analysis project demonstrates an end-to-end data analytics workflow for a highly imbalanced fraud-detection problem. The work covers dataset understanding, exploratory analysis, fraud-pattern investigation, machine-learning modelling, evaluation, and interactive application development.

The Logistic Regression baseline achieved 94.54% overall accuracy and 94% fraud recall on the recorded test results, but the low 2% fraud precision and 4% fraud F1-score show that the baseline is not yet suitable for production-level decision making. The results are valuable because they expose the trade-off clearly and provide a measurable starting point for improvement.

7.2 Future Scope

- Integrate the saved fraud_detection_pipeline.pkl directly into the Streamlit prediction interface.
- Add threshold optimization using validation data so the false-positive rate can be reduced while preserving acceptable fraud recall.
- Evaluate stronger algorithms and ensembles and compare them using precision, recall, F1, PR-AUC, and operational cost.
- Add richer behavioral features such as transaction frequency, repeated sender/receiver patterns, and temporal context.
- Implement monitoring for model drift, changing fraud patterns, and alert volumes.
- Add explainability so an analyst can see which transaction factors contributed to a high-risk prediction.
- Move from a demo environment toward secure, monitored deployment with proper access controls and audit logging.

7.3 Final Project Statement

Final outcome: A documented PaySim fraud-analysis solution that connects data understanding, fraud analytics, machine-learning modelling, evaluation, and an interactive Streamlit presentation layer.



Project Information

Field	Details
Project title	PaySim Fraud Detection & Risk Analysis
Domain	Data Analytics & Artificial Intelligence
Course	Data Analytics
Organization	SURE Trust - IERY
Project member	Amisha Jain
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Project period	July 2026 to September 2026
Primary model	Logistic Regression (class_weight="balanced")
Saved artifact	fraud_detection_pipeline.pkl
Application	Streamlit

Source basis for this report: the supplied SURE Trust - IERY dummy report template, the PaySim Streamlit application files, the fraud-detection pipeline artifact, the supplied ML notebook outputs, and the project READ



Project Github Repository

The complete project source code, notebooks, data files, documentation, and application structure are maintained in a GitHub repository.

GitHub Repository:

<https://github.com/sure-trust/AMISHA-JAIN-g2-data-analytics>

The repository contains the project application, data-processing files, analytical notebooks, documentation, and supporting resources required to understand and reproduce the project.



ME created for this work. No unsupported performance metric has been added.